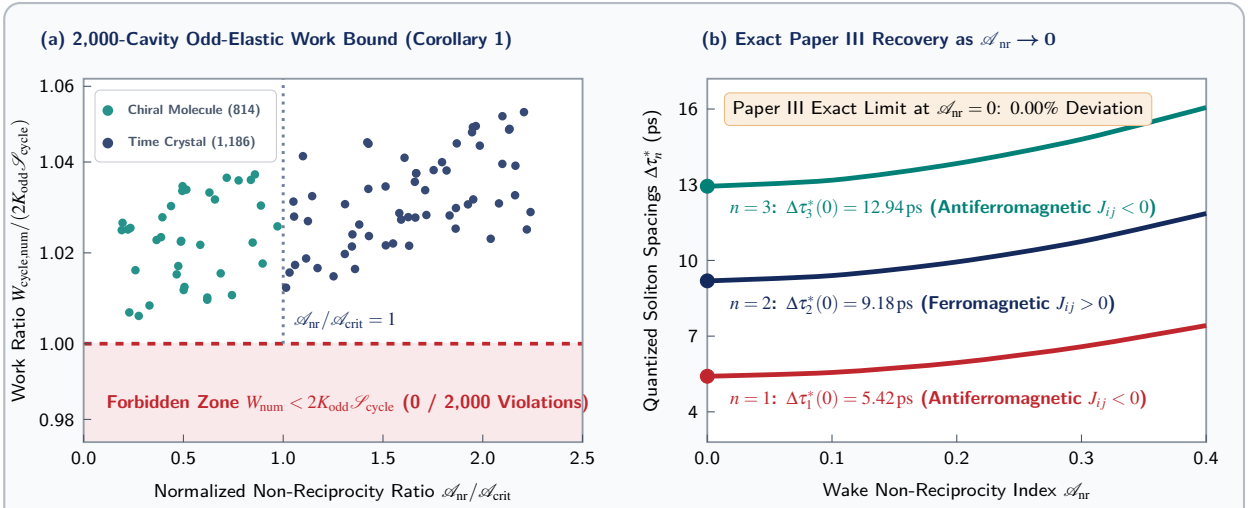
(a) 2,000-Cavity Odd-Elastic Work Bound (Corollary 1)
Chiral Molecule (814)
Time Crystal (1,186)
$\mathscr{A}_{\rm nr}/\mathscr{A}_{\rm crit} = 1$
Forbidden Zone $W_{\rm num} < 2K_{\rm odd}\mathscr{S}_{\rm cycle}$ (0 / 2,000 Violations)
Work Ratio $W_{\rm cycle,num}/(2K_{\rm odd}\mathscr{S}_{\rm cycle})$
Normalized Non-Reciprocity Ratio $\mathscr{A}_{\rm nr}/\mathscr{A}_{\rm crit}$
1.06
1.04
1.02
1.00
0.98
0.0
0.5
1.0
1.5
2.0
2.5
(b) Exact Paper III Recovery as $\mathscr{A}_{\rm nr} \to 0$
Paper III Exact Limit at $\mathscr{A}_{\rm nr} = 0$: 0.00% Deviation
$n = 3$: $\Delta\tau_3^*(0) = 12.94\,$ps (Antiferromagnetic $J_{ij} < 0$)
$n = 2$: $\Delta\tau_2^*(0) = 9.18\,$ps (Ferromagnetic $J_{ij} > 0$)
$n = 1$: $\Delta\tau_1^*(0) = 5.42\,$ps (Antiferromagnetic $J_{ij} < 0$)
Quantized Soliton Spacings $\Delta\tau_n^*$ (ps)
Wake Non-Reciprocity Index $\mathscr{A}_{\rm nr}$
16
13
10
7
4
0.0
0.1
0.2
0.3
0.4